

Lesson 3 · Sketching Gradient Functions

Learning intention

Sketch the graph of $f'(x)$ from the graph of $f(x)$, and sketch a possible $f(x)$ from the graph of $f'(x)$.

Success looks like

- I put the zeros of f' under the turning points of f .
- I put f' on the correct side of the axis in every interval.
- I can go backwards from f' to a possible f .

Slide 1 · Lesson 3 · Sketching Gradient Functions

Teaching line, not a question slide.

Materials: the practice sheet starts with a sketch page; grids 1 to 6 are for the three practice rounds.

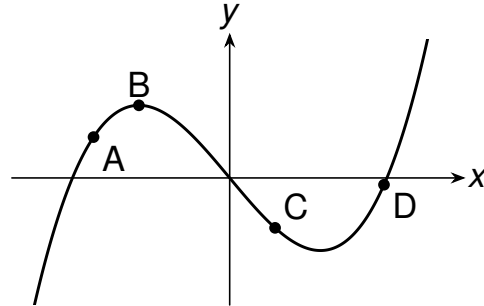
Timing: Do Now 10 min, three rounds about 20 min each, practice sheet 30 min, wrap-up 5 min.

Do Now

1 Find the turning point of $y = x^2 - 6x + 1$ and state its nature.

2 For which x is $f(x) = x^3 - 3x^2$ decreasing?

3 Is the gradient at A, B, C and D positive, negative or zero?



Slide 2 · Do Now

Q1 $(3, -8)$, minimum

Q2 $0 < x < 2$

Q3 A positive; B zero; C negative; D positive

Short solution:

Q1 $2x - 6 = 0$; $y = 9 - 18 + 1$; $y'' = 2 > 0$

Q2 $f'(x) = 3x^2 - 6x = 3x(x - 2) < 0$ between the roots

Q3 $f'(x) = 3x^2 - 3$: 3.75, 0, -2.25, 5.67

Watch for:

Q3 reading the sign of y instead of the direction of the curve (A is below the axis but going up)

Diagnostic:

Q1 or Q2 wrong: Lesson 2 needs a recap before round 2

Q3 wrong: the student confuses height and gradient, the exact confusion this lesson fixes; spend longer on the one idea

The one idea

height of f' = steepness of f

graph of f	graph of f'
turning point	
going up	
going down	
steepest point	

Slide 3 · The one idea

Row 1 on the x -axis ($f' = 0$)

Row 2 above the x -axis

Row 3 below the x -axis

Row 4 turning point of f'

Short solution:

fill the table with the student before any graph; keep it on the whiteboard all lesson

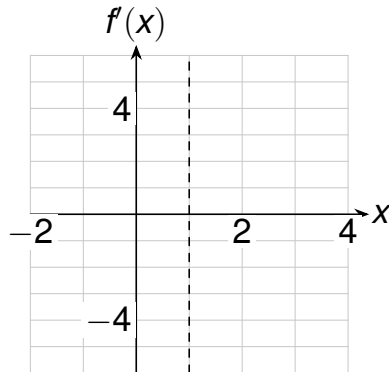
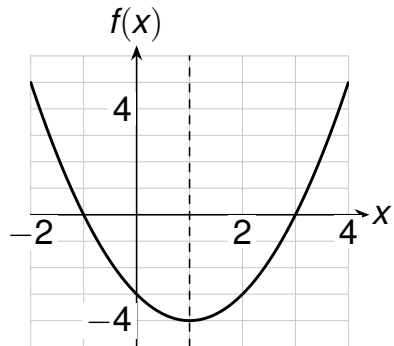
Watch for:

L1: students expect the same hump in the same place; say “flat means zero”

Lines and parabolas

A straight line: f' is a **horizontal line**. A parabola: f' is a **straight line** through the x -value of its turning point.

1 Sketch $f'(x)$ for this parabola, on the axes on the right.



Slide 4 · Lines and parabolas

Q1 straight line crossing the x -axis at $x = 1$, below the axis for $x < 1$, above for $x > 1$

Short solution:

Q1 here $f(x) = x^2 - 2x - 3$, so $f'(x) = 2x - 2$: through $(1, 0)$ and $(0, -2)$

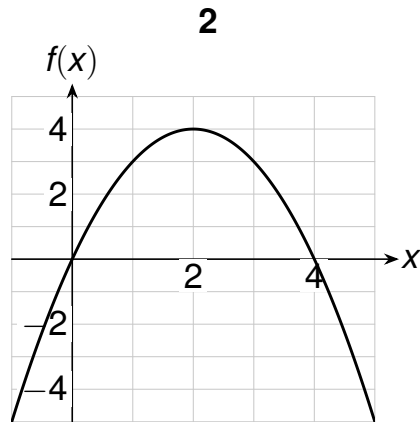
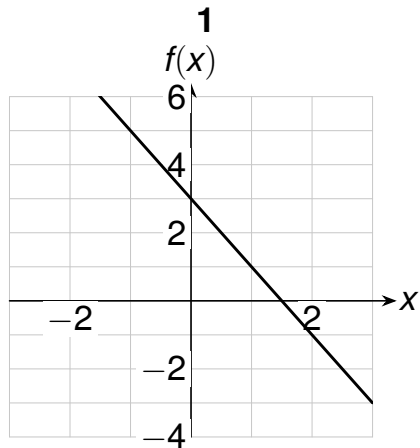
Watch for:

L3: drawing f' as a curve

the dashed line at $x = 1$ is the ruler line: the zero of f' sits on it

Practice

Sketch $f'(x)$ on the sketch page, grids 1 and 2.



Slide 5 · Practice

Q1 horizontal line at height -2

Q2 straight line through $(2, 0)$, above the axis for $x < 2$, below for $x > 2$

Short solution:

Q1 the line falls 2 for every 1 across: gradient -2 everywhere

Q2 maximum at $x = 2$; $f(x) = -x^2 + 4x$, so $f'(x) = -2x + 4$, through $(0, 4)$

Watch for:

Q1 drawing f' as a sloping line

Q2 slope direction: up then down means f' goes from $+$ to $-$

Cubics and quartics

Write it in this order

1. turning points of $f \rightarrow$ zeros of f' (use a ruler).
2. f going up $\rightarrow f'$ above the axis; going down $\rightarrow$ below.
3. Steepest point of $f \rightarrow$ turning point of f' .
4. Join smoothly. Check the type: cubic $\rightarrow$ parabola, quartic $\rightarrow$ cubic.

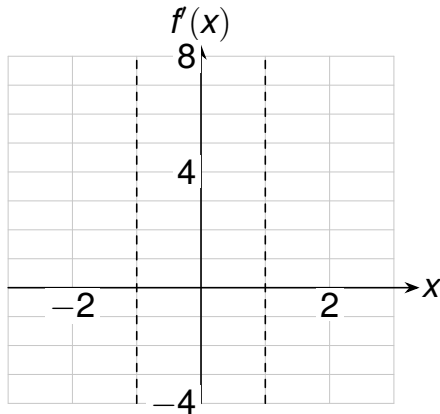
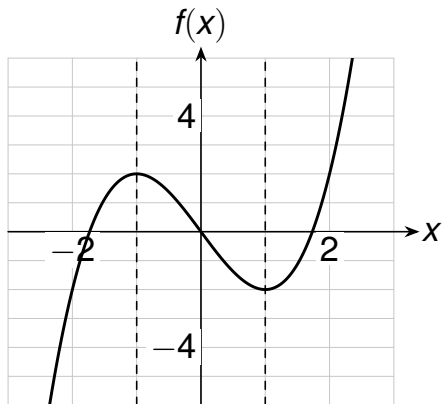
Slide 6 · Cubics and quartics

Teaching line, not a question slide.

Model step 1 with a ruler on the whiteboard: vertical lines from each turning point down to the new axis.

Worked example · cubic

2 Sketch $f'(x)$ for $f(x) = x^3 - 3x$, on the axes on the right.



Slide 7 · Worked example · cubic

Q2 up-parabola with zeros at $x = -1$ and $x = 1$, lowest point at $x = 0$

Short solution:

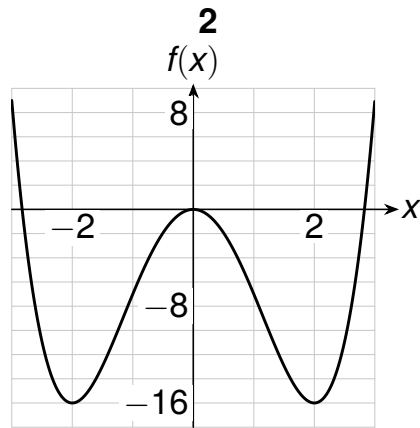
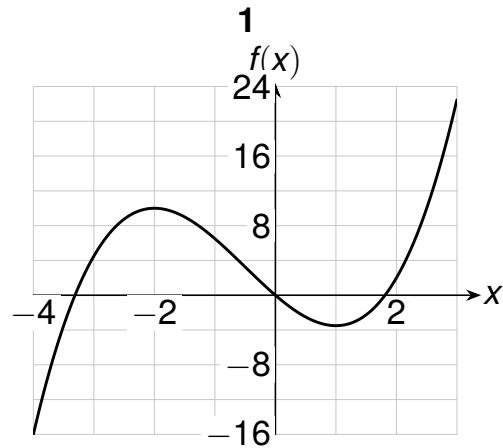
Q2 up, down, up gives above, below, above; $f'(x) = 3x^2 - 3$, minimum -3

Watch for:

L4: the lowest point of f' must be under the steepest downhill part of f ($x = 0$)

Practice

Sketch $f'(x)$ on the sketch page, grids 3 and 4.



Slide 8 · Practice

Q1 up-parabola with zeros at $x = -2$ and $x = 1$

Q2 cubic with zeros at $x = -2, 0$ and 2 : below, above, below, above

Short solution:

Q1 maximum at $x = -2$, minimum at $x = 1$; $f'(x) = 3x^2 + 3x - 6 = 3(x + 2)(x - 1)$

Q2 quartic with minimums at $x = \pm 2$ and a maximum at $x = 0$; $f'(x) = 4x^3 - 16x$

Watch for:

Q1 lowest point of f' at $x = -0.5$, halfway between the zeros

Q2 Merit: a quartic gives a cubic with three zeros

From f' back to f

f' graph	f graph
crosses from $+$ to $-$	
crosses from $-$ to $+$	
above the axis	
below the axis	

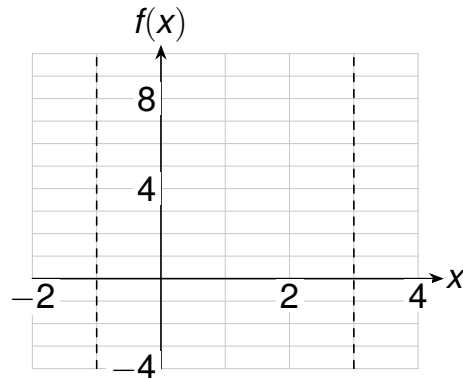
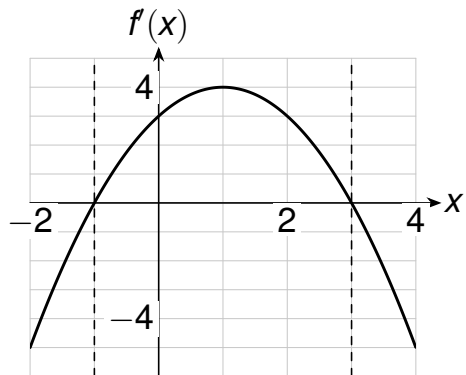
f' gives gradients only, so the height of f is unknown: sketch **a possible f** .

Slide 9 · From f' back to f

- Row 1 maximum
 - Row 2 minimum
 - Row 3 increasing
 - Row 4 decreasing
- Short solution:
read each row from the f' graph left to right
- Watch for:
L5: the zeros of f' are turning points of f , not its x -intercepts

Worked example · backwards

3 Sketch a possible $f(x)$ for this $f'(x)$, on the axes on the right.



Slide 10 · Worked example · backwards

Q3 minimum at $x = -1$, maximum at $x = 3$, decreasing, increasing, decreasing

Short solution:

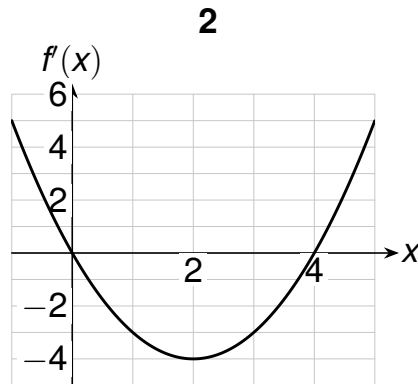
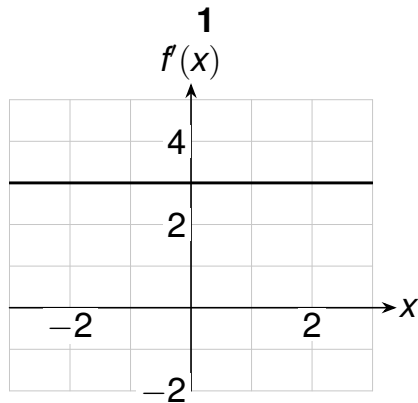
Q3 one possible $f(x) = -\frac{1}{3}x^3 + x^2 + 3x$: minimum $(-1, -1.67)$, maximum $(3, 9)$

Watch for:

L5: drawing f through $x = -1$ and $x = 3$ on the axis

Practice

Sketch a possible $f(x)$ on the sketch page, grids 5 and 6.



Slide 11 · Practice

Q1 any straight line with gradient 3

Q2 maximum at $x = 0$, minimum at $x = 4$

Short solution:

Q1 f' is 3 everywhere, so f rises 3 for every 1 across, e.g. $f(x) = 3x$

Q2 f' goes $+$ to $-$ at 0 and $-$ to $+$ at 4; one possible $f(x) = \frac{1}{3}x^3 - 2x^2 + 4$

Watch for:

Q1 drawing a horizontal line (copying f')

Q2 steepest downhill at $x = 2$, under the lowest point of f'

Ways to lose marks

L1 turning points of f become **zeros** of f' .

L2 f going down $\rightarrow f'$ below the axis.

L3 Parabola $\rightarrow$ line; cubic $\rightarrow$ parabola.

L5 Zeros of f' are turning points of f .

L6 Line up x-values with a ruler; label the graph.

Slide 12 · Ways to lose marks

Teaching line, not a question slide.

Full list L1 to L6 in the handout, with the versions that get no credit.

Summary

1. Height of f' = steepness of f .
2. Turning points of $f \leftrightarrow$ zeros of f' .
3. Up: f' above the axis. Down: f' below.
4. Backwards: $+$ to $-$ is a maximum; the height of f is free.

Slide 13 · Summary

Teaching line, not a question slide.

Ask the student to say each line back before looking at the screen.

Practice sheet · 30 minutes

Work alone on the practice sheet, from Question 1.

- Questions 1 and 2: warm-up, about 6 minutes.
- Questions 3 to 7: exam questions, about 24 minutes.
- Use a ruler to line up every key x -value.

Slide 14 · Practice sheet · 30 minutes

Q1 up-parabola, zeros at $x = -1$ and $x = 3$

Q2 parabola with a maximum at $x = 2$

Q3 cubic: zeros under the three turning points; negative, positive, negative, positive

Q4 cubic shape: minimum under the left zero of f' , maximum under the right zero

Q5 negative quartic shape: maximum at $x = -3$, minimum at $x = -1$, maximum at $x = 3.5$

Q6 (i) $f' = -1$, $g' = 1$, $h'(x) = 2x$

Q7 (ii) $p' = -1$ for $x < 0$ and $p' = 1$ for $x > 0$; (iii) no

Short solution:

Q1 $f'(x) = x^2 - 2x - 3 = (x + 1)(x - 3)$

Q2 f' is $+$ then $-$, crossing at $x = 2$

Q5 f' crosses $+$ to $-$ at -3 and 3.5 , $-$ to $+$ at -1

Q7 (iii) p has a sharp corner at $x = 0$; the gradient jumps from -1 to 1 , so it is not defined there, not zero

Watch for:

Q6 (i) straight lines give horizontal lines; h' passes through the origin

Q7 open circles at $x = 0$ on p'

Full working: practice answer version (tutor's working, not an official mark scheme)

Homework

Homework sheet: 14 questions, about 60 minutes

- Questions 1 to 11: Achieved and Merit sketches.
- Questions 12 and 13: Merit.
- Question 14: Extension (Excellence).

Optional extra: StudyTime Guide, “Sketching Gradient Functions” from page 19.

Slide 15 · Homework

Teaching line, not a question slide.

Answers: homework answer version (tutor’s working, not an official mark scheme).

Next lesson Do Now checks one f to f' sketch in words.